Does CGCSs' Uncertainty about the Evaluator's Identity Affect Performance Differently by Age?

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1. Introduction

In companies and governments, a worker's performance is difficult to measure because assessment requires weighing factors such as quality of work, effort, and contribution that are often hard to compare directly to others (Prendergast 1999; Deb, Li, and Mukherjee 2016). So, many firms and governments consider assessment by supervisors or colleagues as one of the most important components of promotion decisions. However, if the evaluation process rewards influence activities instead of actual performance, it fails to identify who truly deserves promotion. Therefore, masking the evaluator's identity is used as a policy response to reduce this behaviour. This mechanism may extend beyond the Chinese civil service to organizations with multiple evaluators, where concealing the evaluator's identity could similarly reduce targeted lobbying. De Janvry et al. (2023) note that "... when high-stakes rewards are linked to the subjective opinions of designated evaluators, introducing uncertainty in the subjective evaluation scheme could potentially lead to performance improvements" (p. 797). Economists or policymakers should care about this problem because it can happen in any organization that relies on one person's subjective judgement. In programs like CGCS (College Graduate Civil Servants), misdirected promotions waste a large public investment in human capital. As a result, the organization's actual goals are undermined if it cannot distinguish the true contribution from the influence activity.

The main research question is whether the effect of masking the evaluator's identity on CGCSs' performance varies with their age. Since masking (treatment) conceals the evaluator's identity from the CGCS, masking changes the information available to the CGCS, not to the evaluator. Therefore, I interpret the effect of masking on performance as operating through the CGCS's own behavioural response to uncertainty about which supervisor will evaluate them, and this response may vary with age. This paper does not separately identify any changes in the evaluator's own evaluation behaviour. A feature of CGCS program is that selected college graduates enter at different stages of their careers, which can influence how they interact with supervisors and respond to this incentive. However, the data do not contain information on whether any age-related pattern is driven by differences in work experience or in how CGCSs interact with their supervisors. Therefore, I focus on whether the masking effect varies by age, rather than identifying which specific covariates drive it. Age may shape this response because career incentives are strongest in the early stage of a worker's career (Holmström 1999). People become more risk averse over the life course (Dohmen et al. 2017). This could help explain how CGCSs of different ages react to uncertainty about the evaluator's identity.

In the base paper, De Janvry et al. (2023) used separate regressions for the revealed and masked subsamples to show heterogeneity along CGCS – evaluator relationship covariates such as shared hometown or shared party membership. I first test linear heterogeneity using mask x age interaction term. Then, I apply Ridge and LASSO, Decision Tree, ensemble methods (bagging, random forest, and boosting) to consider non-linear perspective that a simple interaction term may miss. My contribution is to examine whether the age heterogeneity exists, using linear, regularized, and non-parametric methods.

The identification strategy is based on masking being randomly assigned at the work-unit level, which means that treatment assignment should be uncorrelated with baseline covariates. Also, I check that attrition does not differ systematically by treatment status. I test these assumptions directly in the paper using a balance table, an attrition test, and additional randomization test.

The average treatment effect is about 0.214, which matches the base paper. Using OLS, Ridge, and LASSO, I checked the robustness of the effect. Despite testing for age heterogeneity using linear and tree-based methods, the results do not clearly show that the masking effect varies by age.

2. Data and Summary Statistics

De Janvry et al. conducted a field experiment among CGCS (College Graduate Civil servants – individual employees who are grouped into work units) in two Chinese provinces. They randomized 3,785 CGCS hired during 2016 and 2017 to either revealed scheme (the evaluator was known at the beginning of the evaluation cycle) or masked scheme (the evaluator remained unknown until the end of the evaluation cycle) for promotion purposes. The experimental design involves random assignment of evaluators and random assignment of masking scheme at the work-unit level.

The baseline survey was conducted in September 2017 which is one month after CGCS completed the job training and received their job assignments. This survey data contains the pre-treatment covariates including age. The endline survey was conducted in June 2018. This survey contains the post-treatment outcome, perf, measured through colleague assessments.

I selected 1 treatment (and its complement), 1 outcome, and 10 baseline covariates: mask (treatment variable), perf (outcome), and 10 baseline covariates (age, female, party member, parent_coll, risk averse, local, social science, college4, stem, village_work). I chose these covariates to check whether the revealed and masked groups are balanced on pre-treatment characteristics.

For the files, the primary data files are the Main survey data and raw data file. The main survey data contains 9,355 peer-evaluation observations, which are aggregated to the CGCS level to construct individual-level performance measures for 2,854 CGCSs. 929 were lost to attrition of the original 3,785 CGCSs between the baseline and endline surveys. After excluding 109 CGCSs with missing age data and two additional CGCSs with missing performance (outcome) data, the final sample in my paper is 2,743 CGCS. Second primary data are attrition data, attrition type data and raw data. These data files contain the records of which CGCS left the sample and the reason why they left during this experiment. These files show that reasons (reassignment and leaving for graduate school or the civil service exam) account for approximately 22.3 percent of the original sample lost to attrition, while overall attrition is 24.5 percent of the same baseline.

Table 1: Summary Statistics of Key Variables (N=2743)

Variable	count	mean	std	min	median	max
mask	2743.000	0.323	0.468	0.000	0.000	1.000
perf	2743.000	5.232	0.911	2.000	5.000	7.000
age	2743.000	24.886	1.622	20.000	25.000	32.000
female	2743.000	0.618	0.486	0.000	1.000	1.000
socsci	2743.000	0.534	0.499	0.000	1.000	1.000
college4	2743.000	0.721	0.448	0.000	1.000	1.000
parent_coll	2743.000	0.290	0.454	0.000	0.000	1.000
stem	2743.000	0.346	0.476	0.000	0.000	1.000
village_work	2743.000	0.154	0.361	0.000	0.000	1.000
risk_averse	2743.000	0.474	0.499	0.000	0.000	1.000
party	2743.000	0.216	0.412	0.000	0.000	1.000
local	2743.000	0.678	0.467	0.000	1.000	1.000

The mean of mask indicates that the proportion of CGCSs assigned to the masked group is close to the

one-third share reported in the original paper. I assess baseline comparability by examining differences in pre-treatment covariates between the masked and revealed groups. Since perf is an ordinal value measured on a 1 to 7 scale, I also estimate an ordered logit/probit models as robustness checks in section 3.7. The mean and median ages are similar, and the relatively small standard deviation suggests a narrow dispersion around mean. The balance tests do not detect statistically significant differences in the other baseline covariates between the masked and revealed groups.

Table 2: Mean Performance (perf) by Age Group and Treatment Status, Full Sample

Age group	Control mean (n)	Treatment mean (n)	Diff. (T−C)	p-value
Young (age $\leq$ 25.5)	5.075 (n=1264)	5.302 (n=587)	0.227	2.66×10^{-7}
Old (age $>$ 25.5)	5.351 (n=592)	5.525 (n=300)	0.174	6.10×10^{-3}

Table 2 splits the full sample into young and old CGCS subgroups using age (25.5) as the cutoff. The estimated masking effect is positive and statistically significant in both groups: 0.227 for younger CGCSs (p ¡ 0.001) and 0.174 for older CGCSs (p = 0.006). The estimated difference between the two effects is 0.053, suggesting a potentially larger effect among younger CGCSs. However, the statistical significance of this difference cannot be inferred from the separate subgroup p-values. This split is descriptive, while formal treatment-effect heterogeneity is examined in Section 3.

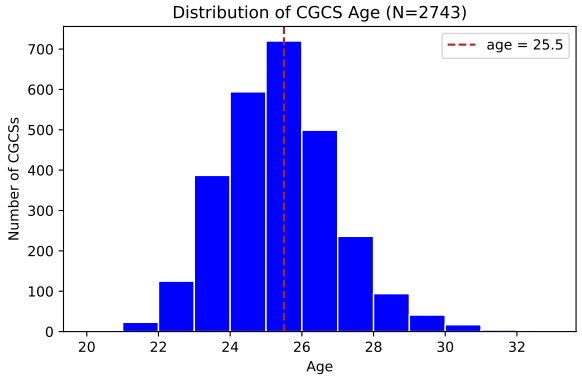


Figure 1: Distribution of CGCS Age(N=2743)

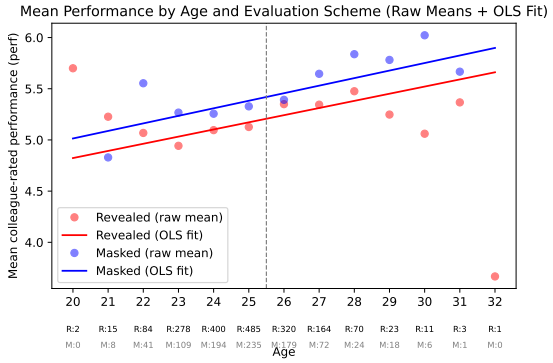


Figure 2: Mean difference between age and evaluation scheme with Sample sizes (N)

In this plot, dashed line (25.5) is the same cutoff as in Figure 1, which I use to split CGCSs into young and old subgroups. The figures to focus on are the N labels beneath each age. This shows how small the sample size gets at certain age categories and how the line can be distorted due to small number of observations. At age 20 and 32, masked N is 0 and at ages between 30 and 31, even revealed N falls to 11 and 3, while ages 23 –26 have 278—485 revealed and 109—235 masked. The line varies widely where N is small (29-32) and is stable where N is large (23-26). Therefore, by looking at the age range 23—26, the gap between masked and revealed does not show a clear pattern, and this means that raw comparison cannot tell whether masking's effect truly varies by age. Therefore, I divide age covariates into several categories and compare estimates with and without fixed effects such as county, CGCS type and cohort.

There are three limitations in this data. First, I dropped 109 CGCS due to missing age. Since all 109

observations are male social-science majors, the final sample may slightly underrepresent this group. Second, the overall attrition rate is relatively high at 24.5 percent. If attrition is related to treatment assignment and potential performance outcomes, the treatment groups in the final sample may no longer be comparable. This potentially biases the estimated effect of masking. I test this formally in the identification strategy section, where attrition does not differ significantly by treatment status. However, similar attrition rates do not mean that the CGCSs who left look the same across the two groups. Third, there is a measurement error in the outcome variable. The outcome perf (colleague assessments) is based on subjective colleague ratings from 1 to 7 rather than an objective performance standard. This means that the estimated effect reflects how colleagues rate performance, not necessarily actual productivity, and masking could even change how colleagues form those ratings. Still, peer evaluation is the actual mechanism this promotion system relies on, so perf remains a meaningful outcome despite this limitation.

3. Regression results

3.1 Identification strategy

The identification strategy in this paper comes from two main components. Masking was randomly assigned at the work-unit level, and attrition does not differ by treatment status.

Table 3: Balance of Baseline Covariates by Treatment Status, Old Subsample (age > 25.5)

Variable	Revealed Mean	Revealed SE	Masked Mean	Masked SE	t	p
female	0.596	0.020	0.570	0.029	0.750	0.453
socsci	0.529	0.021	0.557	0.029	-0.792	0.429
college4	0.887	0.013	0.887	0.018	0.007	0.994
parent coll	0.223	0.017	0.203	0.023	0.680	0.497
stem	0.331	0.019	0.390	0.028	-1.722	0.086
village work	0.147	0.015	0.167	0.022	-0.758	0.449
risk averse	0.456	0.020	0.433	0.029	0.646	0.519
party	0.311	0.019	0.300	0.027	0.331	0.741
local	0.635	0.020	0.663	0.027	-0.836	0.404

Table 4: Balance of Baseline Covariates by Treatment Status, Young Subsample (age ≤ 25.5)

Variable	Revealed Mean	Revealed SE	Masked Mean	Masked SE	t	p
female	0.626	0.014	0.647	0.020	-0.899	0.369
socsci	0.540	0.014	0.514	0.021	1.005	0.315
college4	0.643	0.013	0.639	0.020	0.182	0.856
parent coll	0.324	0.013	0.329	0.019	-0.222	0.824
stem	0.355	0.013	0.317	0.019	1.634	0.102
village work	0.160	0.010	0.141	0.014	1.040	0.299
risk averse	0.479	0.014	0.501	0.021	-0.889	0.374
party	0.170	0.011	0.177	0.016	-0.373	0.709
local	0.701	0.013	0.681	0.019	0.843	0.400

Randomization I divide the total set of CGCS into old and young subgroups. The cutoff is the age (25.5), which I showed in section 2. The two tables (Table 3,Table 4) show that none of nine remaining covariates are different significantly between revealed and masked groups at the 5 percent level. The full sample balance test in Table 11 (**Appendix 1**) also confirms the same pattern.

SUTVA In the base paper, masking is assigned at the work-unit level and 83.9 percent of units contain one CGCS (de Janvry et al. 2023). The remaining 16.1 percent of units have more than one CGCS. All individuals in one work-unit receive the same scheme, so interaction within one group cannot mix revealed and masked outcomes. The assignment of one CGCS is unlikely to affect the performance of another CGCS because evaluators and the reporting system operate within the work units, not across them.

Table 5: Attrition Rate by Treatment Status (t-test)

	Revealed	Masked	t-statistic	p-value
Attrition rate	24.9%	23.9%	-0.647	0.518
N	2575	1210		

No Differential Attrition and Selection Among the total randomized CGCS, 929 of the 3,785 CGCS (24.5 percent) left the sample before the endline survey. The logit regression in Table 12 (**Appendix 2**) shows that attrition ($p = 0.518$) does not differ by treatment status (Table 5). However, a joint logit of attrition on all baseline covariates is significant ($p < 0.001$) (Table 14, **Appendix 3**), so attrition is not completely random for these variables. 109 CGCSs were dropped for missing age and they are all male social-science majors (Table 13, **Appendix 3**). In (Table 13, **Appendix 3**), a logit of missing age variable on treatment status shows no significant relationship ($p = 0.525$).

Through this identification strategy, attrition should not lead to systematic differences between treatment groups for the average treatment effect to be unbiased. Neither attrition nor missing age is significantly associated with treatment status in my tests. However, CGCSs who left the sample may differ systematically from those who stayed. This means the final sample may not represent the full original randomized population. This pattern is also consistent with de Janvry et al. (2023), who report the same attrition rate (24.5 percent) and find that several baseline covariates predict attrition, but attrition does not differ by treatment status.

3.2 DAG

3.2.1 DAG 1 (The Experiment – No Confounding)

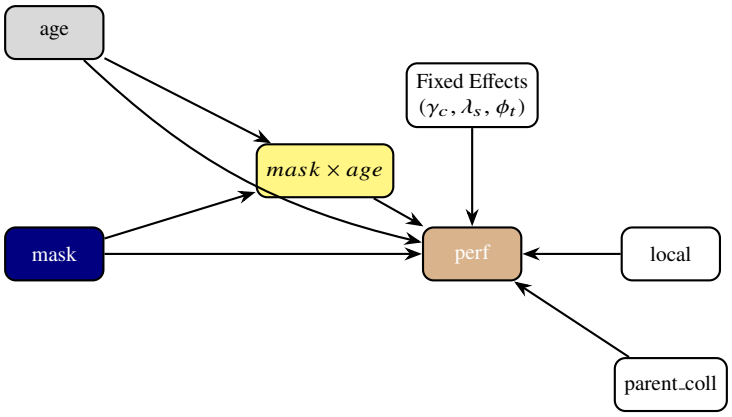


Figure 3: DAG 1: Randomized Treatment Assignment, Fixed Effects, Age, and the Mask×Age Interaction

Due to the randomization of mask (treatment), the other baseline covariates cannot systematically have a confounding relationship between mask and perf. That is, baseline covariates do not create a backdoor path between treatment and outcome. In the DAG 1, this is represented by no arrows from baseline covariates to the treatment assignment. So, the OLS coefficient on mask can be interpreted as an average treatment effect. Mask x age interaction tests whether the masking effect varies with age, rather than removing confounding. Including baseline covariates and fixed effects can improve estimation precision by reducing unexplained variation in the outcome.

3.2.2 DAG 2 – Fork vs. Chain

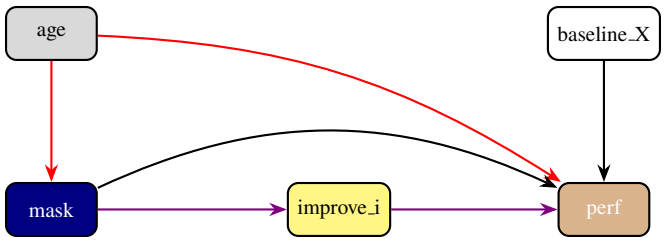


Figure 4: DAG 2: Two Hypothetical Threats – Fork (Age, Red) and Chain (Improve_i, Purple)

Figure 4 illustrates two different causal structures.

In DAG 2, mask (treatment), age, and performance (outcome) are in a fork shape. If age affects both treatment and outcome, it creates a backdoor path and acts as a potential confounder. However, since treatment was randomly assigned, age is not influencing treatment assignment. The balance test in Table 11 also supports this assumption as age does not differ significantly between the treatment groups.

Also, mask (treatment), *improve_i*, and performance (outcome) are in a chain shape and *improve_i* acts as a mediator. Controlling for this mediator would block the indirect effect and prevent estimation of the total treatment effect. Therefore, *improve_i* should not be controlled.

3.2.3 DAG 3 – Collider (Sample Attrition)

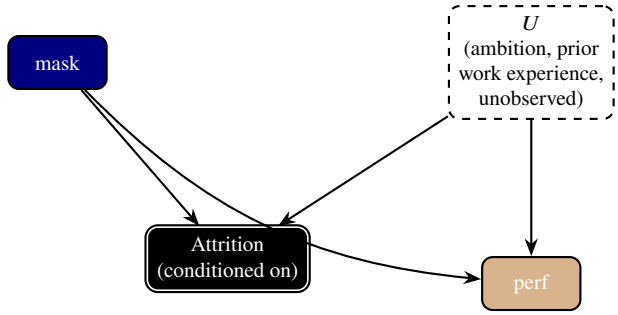


Figure 5: DAG 3: Potential Collider Bias from Conditioning on Sample Attrition

Figure 5 illustrates a potential collider problem arising from the sample attrition. In DAG 3, mask (treatment), attrition, and the unobserved factor are in a collider shape. If I control for attrition, it creates a spurious relationship between mask and the unobserved variable. This could bias the estimated causal effect of mask on perf. So, I test whether mask is associated with attrition and the logit regression in Table 12 finds no statistical association between mask and attrition (p = 0.518). However, this does not guarantee that mask has no effect on attrition, so selection bias remains a potential limitation of this paper.

3.3 Linear regression - OLS

$$\begin{aligned} perf_{icst} = {} & \beta_1 \text{mask}_{icst} + \beta_2 \text{age}_{icst} + \beta_3 \text{female}_{icst} \\ & + \beta_4 \text{socsci}_{icst} + \beta_5 \text{college4}_{icst} + \beta_6 \text{parent_coll}_{icst} \\ & + \beta_7 \text{stem}_{icst} + \beta_8 \text{village_work}_{icst} + \beta_9 \text{risk_averse}_{icst} \\ & + \beta_{10} \text{party}_{icst} + \beta_{11} \text{local}_{icst} + \gamma_c + \lambda_s + \phi_t + \epsilon_{icst} \end{aligned}$$

where $\gamma_c + \lambda_s + \phi_t$ are county, CGCS-type, and cohort fixed effects respectively.

Table 6: OLS Regression Results

	Coefficient	Std. Err.	z	p-value	95% CI
const	0.0000	0.016	0.000	1.000	[-0.031, 0.031]
mask	0.2136	0.035	6.150	0.000	[0.146, 0.282]
age	-0.0204	0.012	-1.636	0.102	[-0.045, 0.004]
female	-0.0029	0.036	-0.082	0.935	[-0.073, 0.067]
socsci	0.0502	0.036	1.388	0.165	[-0.021, 0.121]
college4	0.0120	0.044	0.271	0.786	[-0.075, 0.098]
parent_coll	-0.0834	0.037	-2.255	0.024	[-0.156, -0.011]
stem	0.0187	0.037	0.506	0.613	[-0.054, 0.091]
village_work	0.0805	0.049	1.629	0.103	[-0.016, 0.177]
risk_averse	-0.0219	0.033	-0.672	0.502	[-0.086, 0.042]
party	0.0562	0.040	1.399	0.162	[-0.023, 0.135]
local	0.1128	0.040	2.833	0.005	[0.035, 0.191]

N = 2,743; R^2 = 0.022; Adj. R^2 = 0.018; F-statistic = 5.840 ($p < 0.001$). Standard errors clustered at the work-unit level.

This OLS regression estimates the average treatment effect of mask on performance, while controlling for age and other baseline covariates. The mask (treatment) has a coefficient of 0.2136, which is significant at

the 5 percent level (z = 6.150, p < 0.001). Parent_coll (coefficient = -0.0834, p = 0.024) and local (coefficient = 0.1128, p = 0.005) covariates are significant at the 5 percent level. The R^2 of 0.022 reflects only the explanatory power of the baseline covariates after removing the variation explained by the fixed effects.

3.4 Linear regression – OLS using Mask–age interaction term

$$perf_{icst} = \beta_1 \text{mask}_{icst} + \beta_2 \text{age}_{icst} + \beta_3 (\text{mask}_{icst} \times \text{age}_{icst}) + \text{controls}_{icst} + \gamma_c + \lambda_s + \phi_t + \epsilon_{icst}$$

where age is centered at its sample mean, controls include the same baseline covariates as in Section 3.3, and $\gamma_c + \lambda_s + \phi_t$ are county, CGCS-type, and cohort fixed effects.

Table 7: Mask × Age interaction. Dependent variable: average colleague assessment (1–7). N = 2743, R^2 = 0.022. Age is centered at its sample mean; all other baseline covariates, missing-value indicators, and county/CGCS-type/cohort fixed effects are included but omitted from the table. Standard errors clustered at the work-unit level.

	Coefficient	Std. Error	t-Statistic	p-Value	95% CI low	95% CI high
Masked scheme	0.2137	0.0347	6.1497	0.0000	0.1456	0.2817
Age (centered)	-0.0214	0.0143	-1.4962	0.1346	-0.0494	0.0066
Masked scheme × Age (centered)	0.0031	0.0219	0.1395	0.8890	-0.0399	0.0461

The OLS regression with mask x age interaction added estimates whether the masking effect linearly varies with age. The coefficient on the mask is 0.2137 (p ¡ 0.001), which is nearly identical to the baseline OLS estimate of 0.2136. However, the coefficient on mask x age interaction is 0.0031 (p = 0.889) and this is statistically insignificant at the 5 percent level. This shows that the estimated linear relationship between age and the masking effect is nearly zero. Therefore, I found no linear age heterogeneity patterns through this analysis. Age is not randomly distributed across the baseline covariates (Table 15, Appendix 4).

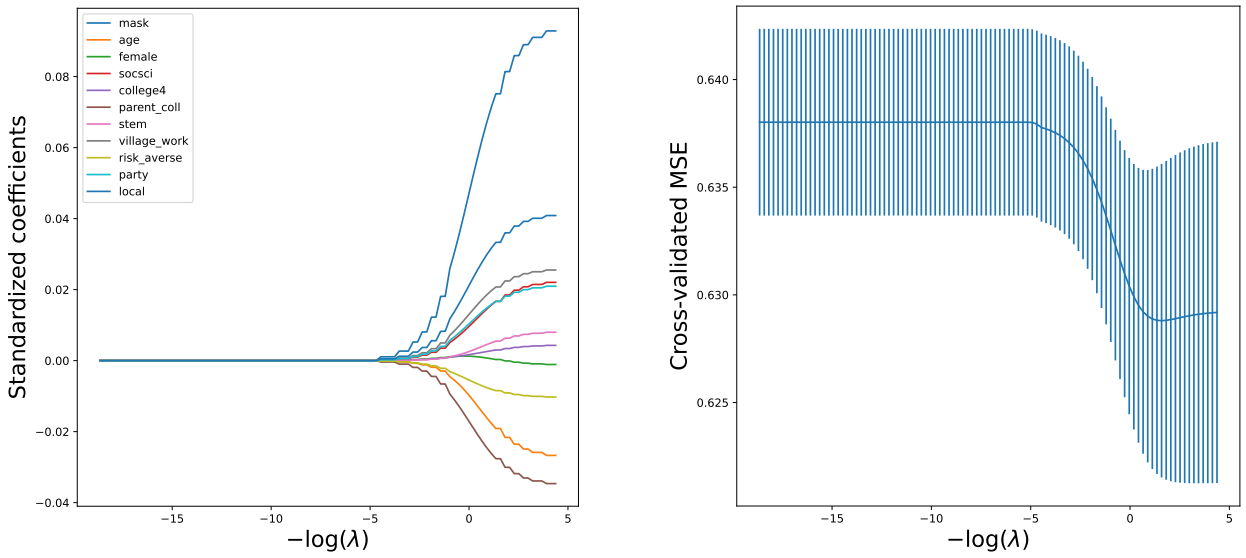
3.5 Ridge and LASSO

3.5.1 Ridge

$$\hat{\beta}^{Ridge} = \arg\min_{\beta} \; RSS + \lambda \sum_{j} \beta_j^2 \quad (\ell_2)$$

where $\lambda \geq 0$, with $\lambda = 0$ reducing to OLS; the intercept is not penalized, and covariates are standardized before estimation.

$$RSS = \sum_{i} \left(\widetilde{perf}_{icst} - \beta_1 \widetilde{\text{mask}}_{icst} - \beta_2 \widetilde{\text{age}}_{icst} - \cdots - \beta_{11} \widetilde{\text{local}}_{icst} \right)^2$$



(a) coefficient paths

(b) Cross-validated MSE

Figure 6: Ridge regression: standard coefficient (left) and 5-fold cross-validated MSE (Right) across values of $-log(\lambda)$. Fixed effects are partialled out beforehand. Dotted line implies the CV-minimizing λ

Ridge estimates a masked-scheme effect of 0.189, which is similar to the OLS estimate of 0.2136. Also, Ridge has a minimum cross-validated MSE of 0.6288. Since Ridge's penalty is quadratic, small coefficients shrink toward zero but never reach zero. Therefore, Ridge keeps every covariate in the model.

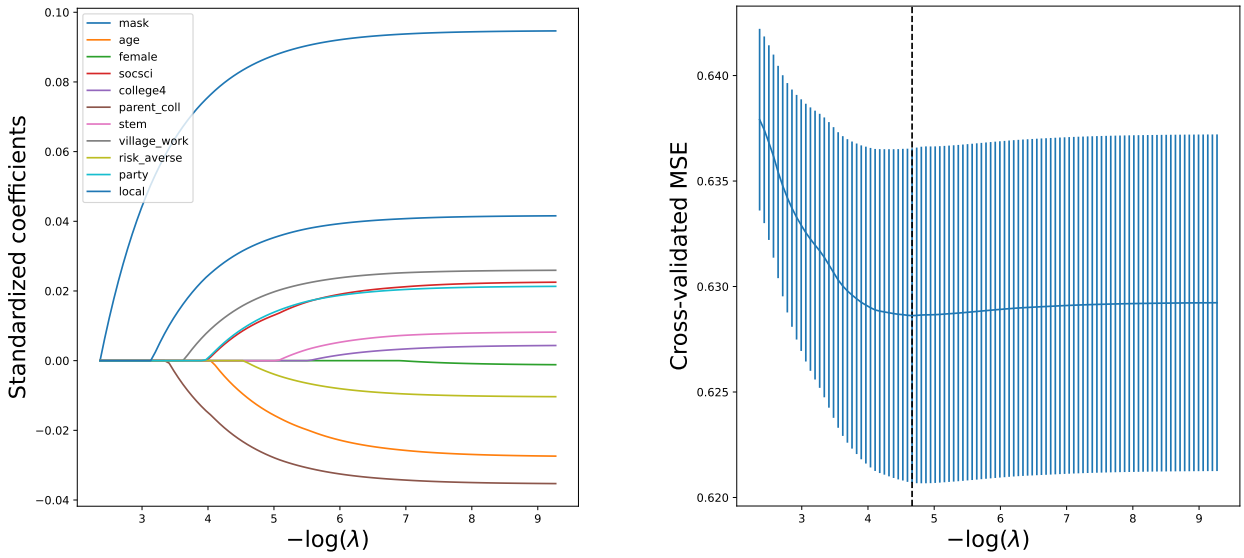
3.5.2 LASSO

$$\hat{\beta}^{LASSO} = \arg\min_{\beta} \; RSS + \lambda \sum_{j} |\beta_j| \quad (\ell_1)$$

where $\lambda \geq 0$, with $\lambda = 0$ reducing to OLS; the intercept is not penalized, and covariates are standardized before estimation.

$$RSS = \sum_{i} \left(\widetilde{perf}_{icst} - \beta_1 \widetilde{\text{mask}}_{icst} - \beta_2 \widetilde{\text{age}}_{icst} - \cdots - \beta_{11} \widetilde{\text{local}}_{icst} \right)^2$$

where the tilde denotes each variable residualized against fixed effects via Frisch-Waugh-Lovell theorem, and λ is selected by 5-fold cross-validation. Unlike Ridge, the ℓ_1 penalty can shrink coefficient to zero.



(a) Coeffcieint paths

(b) Cross-validated MSE

Figure 7: LASSO regression: standard coefficient (left) and 5-fold cross-validated MSE (Right) across values of $-log(\lambda)$. Fixed effects are partialled out beforehand. Dotted line implies the CV-minimizing λ

LASSO estimates a masked-scheme effect of 0.191, which is similar to the OLS estimate of 0.2136. Also, LASSO has a minimum cross-validated MSE of 0.6286. This is nearly identical to Ridge's. Figure 7(b) shows the cross-validated MSE curve reaching its minimum at a similar $-\log(\lambda)$ to Ridge's. Unlike Ridge, LASSO can shrink small coefficients to exactly zero. Therefore, it drops female, college4, and stem, which had the highest p-values under OLS. Figure 7(a) shows a similar pattern to Ridge's, except that the dropped covariates' coefficients converge to zero. These paths graphically show LASSO's variable-selection process.

A null model with no covariates has a cross-validated MSE of 0.6380, so both Ridge and LASSO improve only by 1.4 and 1.5 percent respectively. Although LASSO dropped three variables, LASSO and Ridge methods achieve nearly identical predictive accuracy. This implies the three dropped variables contribute little to predicting performance (outcome). Also, compared to OLS, both methods shrink the mask coefficient (treatment) slightly, but it still remains the largest coefficient in both paths, which means it is consistent with the OLS results.

3.6 Decision Tree and Ensemble methods

3.6.1 Decision Tree

$$\sum_m RSS_m + \alpha|T|$$

where $|T|$ is the number of terminal nodes and α plays the same role as λ in LASSO, trading off fit against tree complexity.

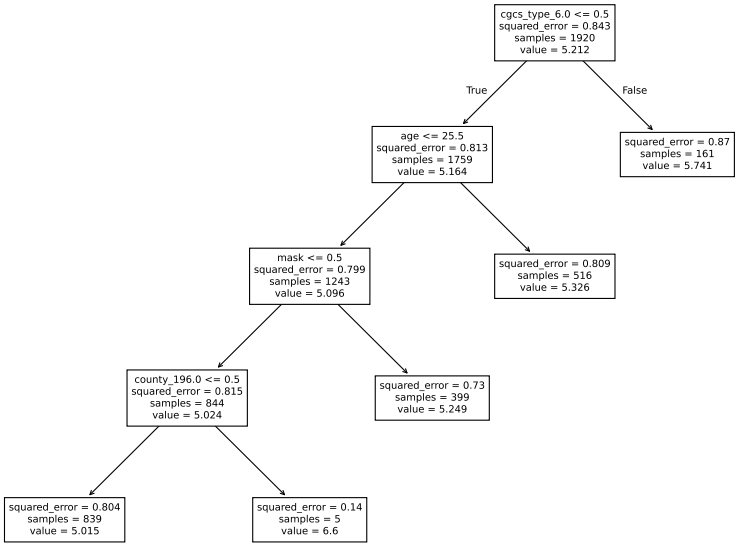


Figure 8: Pruned decision tree predicting perf

The decision tree estimates a masked-scheme effect separately for each subgroup defined by its splits to capture non-linear and interaction effects that the OLS may miss. The root split is on cgcs_type_6, which is a CGCS-type fixed effect dummy. The second split occurs at age 25.5. Below age 25.5, the tree splits further on mask. The revealed branch has an average of 5.02, while the masked branch has an average of 5.25 with no further split. The gap is about 0.23. Above age 25.5, the tree makes no further split and ends in a single leaf. The pruned tree with 5 leaves has a test MSE of 0.779.

Also, I compare the decision tree and OLS. OLS has no mask-by-age interaction in the specification, so it assumes the same masking effect (0.2136) across all ages. The root split on cgcs_type_6 rather than mask or age suggests that CGCS type contributes more to the initial reduction in prediction error than splits based on treatment status or age. The age split at 25.5 reflects where age best reduces prediction error, not a formally estimated heterogeneous treatment effect.

3.6.2 Bagging

$$\hat{f}_{bag}(x) = \frac{1}{B}\sum_{b=1}^{B} \hat{f}^{*b}(x)$$

Unlike the decision tree, bagging averages predictions across many bootstrapped trees. For bagging, I used 300 trees and tuned the maximum depth to 8. Bagging reduces the test MSE to 0.749, which is slightly lower than the single pruned tree (0.779). This is consistent with variance reduction through bootstrap aggregation.

3.6.3 Random Forest

$$\hat{f}_{RF}(x) = \frac{1}{B}\sum_{b=1}^{B} \hat{f}^{*b}(x)$$

Table 8: Random Forest Variable Importance (Top 15)

Variable	Importance
age	0.0918
risk_averse	0.0345
socsci	0.0316
female	0.0315
stem	0.0306
parent_coll	0.0303
local	0.0267
year_2017	0.0267
mask	0.0246
cgcs_type_6.0	0.0240
party	0.0217
village_work	0.0213
college4	0.0212
cgcs_type_2.0	0.0209
cgcs_type_4.0	0.0156

Unlike bagging, random forest reduces correlation between trees by selecting predictors. The tuned random forest with unrestricted depth and max_features = 11 achieves a test MSE of 0.748, close to bagging's 0.749. The most influential feature is age (0.092) and its importance is almost three times bigger than that of the second most influential feature, risk_averse (0.0345). These findings are consistent with the decision tree in which age was the first variable selected after the split on cgcs_type_6. Random forest variable importance measures each variable's contribution to prediction error reduction, not the size of the treatment effect, so the low contribution of mask does not mean the masking effect is small.

3.6.4 Boosting

$$\hat{f}(x) = \sum_{b=1}^{B} \lambda \hat{f}^b(x)$$

where $Y = \lambda\hat{f}_1(x) + \lambda\hat{f}_2(x) + \cdots + \lambda\hat{f}_B(x) + r_B$: each tree is fit to the residual from the previous stage, and λ is the shrinkage (learning rate); in practice λ is small (e.g., 0.01 or 0.001) and B is large.

Table 9: Test-Set Prediction Error by Model

Model	Test MSE	Test RMSE
Single Tree (pruned)	0.779	0.883
Bagging	0.749	0.866
Random Forest	0.748	0.865
Boosting	0.730	0.854

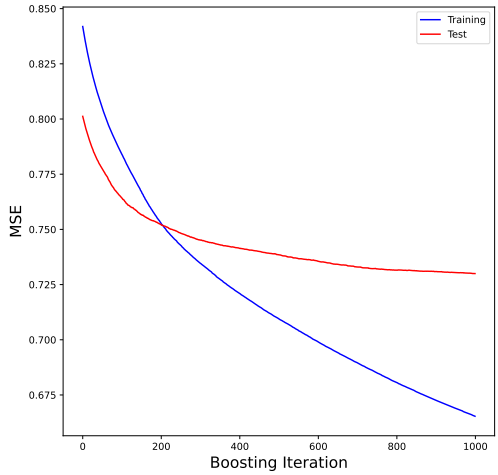


Figure 9: Boosting training and test MSE by iteration.

Unlike bagging and random forest, boosting fits each tree sequentially to the residuals of the previous trees, which can explain its lower test error. Boosting (learning rate = 0.01, max_depth = 3) achieves a test MSE of 0.730, which is the lowest test MSE among the four models. This is an improvement over both bagging (0.749) and the tuned random forest (0.748).

Overall, the tree-based methods have test MSE in the range of 0.730 to 0.779.

3.7 Logit and Probit

The performance (outcome) is recorded on an ordinal 1–7 scale. Therefore, I estimate ordered logit and probit models as robustness checks. Through these models, I test whether the estimated effect of masking remains positive and statistically significant.

$$Y_i^* = \alpha \times Mask_i + X_i'\gamma + \lambda_s + \phi_t + \epsilon_i, \tag{1}$$

where Y_i^* is the unobserved latent performance propensity, and Y_i is the observed ordinal performance score ranging from 1 to 7. α is the coefficient on the treatment indicator $Mask_i$, and γ is the vector of coefficients on X_i', which contains the same baseline covariates as in Section 3.3. λ_s and ϕ_t denote CGCS-type and cohort fixed effects, respectively. ϵ_i follows a logistic distribution under ordered logit and a standard normal distribution under ordered probit.

Table 10: Effect of Masking on Performance (Ordered Logit and Probit, Individual-Level, Cluster-Robust SE at Work-Unit Level)

Model	Coefficient	Std. Err.	z	p-value
Ordered Logit	0.3439	0.0531	6.473	0.000
Ordered Probit	0.1959	0.0315	6.217	0.000

The ordered probit coefficient (0.1959) is smaller than the ordered logit coefficient (0.3439), and both of them are significant at the 1 percent level. Both estimates are positive, and this is consistent with the OLS mask coefficient (0.2136). This is consistent with the standard logit-probit ratio (1.6–1.8).

4. Conclusion

In conclusion, this paper studies whether CGCSs' uncertainty about the evaluator's identity affects performance, and whether this effect differs by age. Masking is consistently associated with better performance across every specification. This is also consistent with the positive average treatment effect found in de Janvry et al. (2023). The simple linear interaction model finds no statistically significant evidence of age heterogeneity. Based on the results in ensemble methods, the findings suggest that age is relevant for predicting performance, but they do not confirm that the causal effect of masking varies by age. Overall, these results leave the research question only partially resolved.

5. Reference

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6. Appendix

Appendix 1: Balanced Table

Table 11: Balance of Baseline Covariates by Treatment Status

Variable	Revealed Mean	Revealed SE	Masked Mean	Masked SE	t	p
age	24.866	0.038	24.928	0.054	0.934	0.351
female	0.616	0.011	0.621	0.016	0.243	0.808
socsci	0.536	0.012	0.529	0.017	-0.361	0.718
college4	0.721	0.010	0.723	0.015	0.096	0.924
parent_coll	0.291	0.011	0.286	0.015	-0.277	0.782
stem	0.348	0.011	0.342	0.016	-0.305	0.760
village_work	0.156	0.008	0.150	0.012	-0.394	0.694
risk_averse	0.471	0.012	0.478	0.017	0.322	0.747
party	0.215	0.010	0.219	0.014	0.222	0.825
local	0.680	0.011	0.675	0.016	-0.243	0.808

This is combined balance table across full sample (Not split by young and old subgroups)

Appendix 2: Attrition and Randomization Diagnostics

Table 12: Logit Regression of Attrition on Treatment Status

	Coefficient	Std. Err.	z	p-value
mask	-0.0526	0.0814	-0.647	0.518

The logit coefficient on treatment status is small and statistically insignificant (p = 0.518), consistent with the balance table (Table 11) in Section 3.1 and the attrition t-test (Table 5): attrition is not systematically related to masking status.

Appendix 3: Missing Data and Propensity Score Diagnostics

Table 13: Logit Regression of Missing-Age Indicator on Baseline Covariates (Excluding Female, Social Science)

	Coefficient	Std. Err.	z	p-value
mask	-0.1364	0.2146	-0.636	0.525
college_4	0.1937	0.2336	0.829	0.407
parent_college	-0.3113	0.2325	-1.339	0.181
stem	-0.0647	0.2077	-0.311	0.755
village_work	0.4306	0.2384	1.807	0.071
risk_averse	-0.0295	0.1967	-0.150	0.881
party_member	0.1254	0.2307	0.544	0.587
local	0.3990	0.2290	1.743	0.081

Excluding female and social science, which perfectly predict the missing-age indicator (all 109 cases are male social-science majors), a logit on the remaining baseline covariates shows no significant joint relationship (LR p = 0.258), and treatment status does not predict missingness (p = 0.525).

Table 14: Joint Predictiveness of Baseline Covariates for Treatment and Attrition

	Joint LR χ^2 p-value	Pseudo R^2
P(T=mask | X)	0.143	0.0028
P(A=attrite | X, T)	0.0000	0.0091

A joint logit of treatment status on all ten baseline covariates is insignificant (p = 0.143), which strengthen the balance result in section 3.1. A joint logit of attrition on the same covariates plus treatment status is highly significant (p < 0.001), indicating that the baseline covariates jointly predict attrition.

Appendix 4: OLS regression - Age

Table 15: OLS Regression of Age on Baseline Covariates

Variable	Coefficient	Std. Err.	z	p-value
female	-0.2428	0.0609	-3.986	0.000
socsci	0.1519	0.0600	2.532	0.011
college4	1.2416	0.0618	20.091	0.000
parent_coll	-0.4984	0.0599	-8.317	0.000
stem	-0.0248	0.0636	-0.390	0.696
village_work	-0.0655	0.0794	-0.824	0.410
risk_averse	-0.1228	0.0562	-2.185	0.029
party	0.6071	0.0752	8.075	0.000
local	-0.1283	0.0616	-2.082	0.037

Age is predicted jointly by the other baseline covariates ($F = 64.85$, $p < 0.001$), with college4, party, parent_coll, and female as the strongest predictors.